

Design, Theoretical Analysis, and Simulation of a 3-Stage Current-Starved Voltage-Controlled Ring Oscillator in 90nm CMOS Technology

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Abstract—This paper presents the exhaustive theoretical derivation, architectural design, and simulation analysis of a 3-stage Current-Starved Voltage-Controlled Ring Oscillator (VCO) implemented in a standard 90 nm Analog CMOS process under a 1.2 V supply voltage. The fundamental principles of phase-shift generation, Barkhausen oscillation criteria, small-signal loop gain, and large-signal propagation delay τ_d are rigorously derived. Frequency tuning is achieved by modulating the charging and discharging current I_{ctrl} of the inverter delay cells via an applied control voltage V_{ctrl} . The mathematical relation $f_{\text{osc}} = \frac{1}{2N\tau_d} = \frac{I_{\text{ctrl}}}{2NC_L V_{\text{DD}}}$ and the corresponding VCO sensitivity gain $K_{\text{VCO}} = \frac{\partial f_{\text{osc}}}{\partial V_{\text{ctrl}}}$ are derived from MOSFET square-law device physics. Simulation results extracted from Cadence Virtuoso sweeps across control voltages from 0.5 V to 1.2 V demonstrate an output frequency tuning range from 3.79 GHz at $V_{\text{ctrl1}} = 0.5$ V ($f_{\text{osc1}} = 3788$ MHz), 8.09 GHz at $V_{\text{ctrl2}} = 0.8$ V ($f_{\text{osc2}} = 8094$ MHz), to 9.75 GHz at $V_{\text{ctrl3}} = 1.2$ V ($f_{\text{osc3}} = 9746$ MHz). The design achieves an exceptionally wide continuous tuning bandwidth of 5.96 GHz with an average K_{VCO} of 8511 MHz/V (8.51 GHz/V) and a total power consumption of 1.26 mW at $V_{\text{DD}} = 1.2$ V. Performance metrics are systematically compared against state-of-the-art literature.

Index Terms—Analog CMOS Integrated Circuits, Voltage-Controlled Oscillator (VCO), Current-Starved Ring Oscillator, Barkhausen Criteria, Propagation Delay, Tuning Range, K_{VCO} , 90nm CMOS.

I. INTRODUCTION AND LITERATURE SURVEY

Voltage-Controlled Oscillators (VCOs) are essential components in contemporary integrated analog, mixed-signal, and radio-frequency (RF) communication systems. They function as the core frequency synthesis unit in Phase-Locked Loops (PLLs), Clock and Data Recovery (CDR) circuits, local oscillators for wireless transceivers, and networks for generating high-speed digital clocks. The main purpose of a VCO is to transform an analog control voltage V_{ctrl} into a periodic output waveform, where the instantaneous oscillation frequency f_{osc} varies monotonically with V_{ctrl} .

In the realm of Analog CMOS Integrated Circuits, two primary VCO topology categories are extensively utilized: *LC-tank oscillators* and *ring oscillators*. LC oscillators are known for their excellent phase noise characteristics and reduced high-frequency jitter. However, the integrated spiral inductors associated with them demand significant silicon area, have a

restricted quality factor (Q), and offer limited frequency tuning ranges (up to 15%). On the other hand, CMOS Ring Oscillators present significant architectural benefits:

- 1) **Compact Layout Footprint:** The removal of large passive inductors and transformers allows for monolithic integration within cost-effective digital CMOS processes.
- 2) **Wide Frequency Tuning Range:** Achieving a broad frequency tuning range (exceeding 100%) is possible through tail current modulation or by adjusting capacitive loading.
- 3) **Multiphase Output Generation:** Evenly spaced multiphase clock outputs ($\phi_k = \frac{k\pi}{N}$) are naturally available, which are essential for quadrature modulation and clock recovery.

The latest research encompasses various types of ring oscillators, such as traditional single-ended inverter rings, differential delay-cell rings, cross-coupled latch rings, and current-starved ring configurations. Notably, the **current-starved ring VCO** is distinguished by its broad linear voltage-to-frequency tuning range, ability to achieve full rail-to-rail voltage swings, and straightforward current mirror biasing.

II. LIMITATIONS OR DRAWBACKS OF CURRENTLY AVAILABLE TECHNOLOGY

Non-tunable ring oscillators are typically made up of an odd number N of fundamental CMOS inverter stages arranged in a closed feedback loop. Despite their simplicity, these basic uncontrolled ring oscillators have significant limitations when used in practical analog CMOS integrated circuits:

- 1) **Absence of Frequency Control:** The oscillation frequency is fixed entirely by the intrinsic propagation delays of the inverters ($\tau_d \approx R_{\text{on}}C_L$), making electronic tuning impossible.
- 2) **High Sensitivity to Process, Voltage, and Temperature (PVT) Variations:** Fabrication tolerances in threshold voltage V_{TH} , oxide thickness t_{ox} , and supply noise directly fluctuate the oscillation frequency by up to $\pm 35\%$.
- 3) **High Power Dissipation at Uncontrolled Speeds:** Unrestricted charging currents result in excessive dynamic power consumption $P_{\text{dyn}} = NC_L V_{\text{DD}}^2 f_{\text{osc}}$.

To overcome these fundamental limitations, the current-starved architecture introduces voltage-controlled current-sinking and current-sourcing transistors in series with each inverter delay stage, thereby decoupling the frequency setting from raw supply rail fluctuations and enabling a wide-range electronic control.

III. PROPOSED SOLUTION AND METHODOLOGY

A. Circuit Architecture and Delay Cell Operation

The proposed 3-stage Current-Starved Voltage-Controlled Ring Oscillator consists of three identical inverting delay stages connected in a ring topology with a global current mirror control network.

Each delay cell consists of a standard CMOS inverter pair (M_{P1} – M_{N1}) augmented with two current-limiting "starving" transistors: a PMOS tail transistor M_{CP} connected to V_{DD} and an NMOS tail transistor M_{CN} connected to ground. The gate voltage of M_{CN} is directly controlled by the external control voltage V_{ctrl} , while $V_{ctrl,p}$ for M_{CP} is mirrored via a PMOS diode-connected reference branch to ensure $I_{source} = I_{sink} = I_{ctrl}$.

When the input to a stage transitions High-to-Low, the PMOS M_{P1} turns ON and charges the load capacitance C_L through M_{CP} with current I_{ctrl} . Conversely, when the input transitions Low-to-High, NMOS M_{N1} turns ON and discharges C_L to Ground through M_{CN} with current I_{ctrl} .

B. Barkhausen Criteria and Small-Signal Phase Analysis

For steady-state sustained oscillations to occur in any linear or weakly non-linear feedback system, the loop transfer function $H(s) = A(s) \cdot \beta(s)$ must satisfy the *Barkhausen Criteria*:

$$|H(j\omega_0)| \geq 1 \quad (1)$$

$$\angle H(j\omega_0) = 180^\circ + 360^\circ \times k \quad (k \in \mathbb{Z}) \quad (2)$$

For an N -stage ring oscillator consisting of identical single-pole inverting stages, each stage exhibits a small-signal transfer function:

$$A_i(s) = \frac{-A_0}{1 + \frac{s}{\omega_p}} \quad (3)$$

where $A_0 = g_m(r_{oN} \parallel r_{oP})$ is the small-signal DC voltage gain per stage, $\omega_p = \frac{1}{R_{out}C_L}$ is the dominant pole frequency, and the negative sign represents the 180° low-frequency phase inversion inherent to the CMOS inverter.

The total open-loop transfer function for $N = 3$ stages is:

$$A_{loop}(s) = \left(\frac{-A_0}{1 + \frac{s}{\omega_p}} \right)^3 = \frac{-A_0^3}{\left(1 + \frac{s}{\omega_p} \right)^3} \quad (4)$$

At the oscillation frequency $\omega_0 = 2\pi f_{osc}$, the phase shift contributed by the frequency-dependent poles across all 3 stages must equal 180° (so that together with the intrinsic odd-number inversions, the total loop phase shift equals 360°):

$$3 \times \arctan\left(\frac{\omega_0}{\omega_p}\right) = 180^\circ \implies \arctan\left(\frac{\omega_0}{\omega_p}\right) = 60^\circ = \frac{\pi}{3} \text{ rad} \quad (5)$$

Solving Eq. (5) for the small-signal oscillation frequency ω_0 :

$$\frac{\omega_0}{\omega_p} = \tan(60^\circ) = \sqrt{3} \implies \omega_0 = \sqrt{3}\omega_p = \frac{\sqrt{3}}{R_{out}C_L} \quad (6)$$

Substituting ω_0 into the magnitude criterion $|A_{loop}(j\omega_0)| \geq 1$:

$$\frac{A_0^3}{\left[\sqrt{1 + \left(\frac{\omega_0}{\omega_p} \right)^2} \right]^3} = \frac{A_0^3}{[\sqrt{1+3}]^3} = \frac{A_0^3}{8} \geq 1 \quad (7)$$

$$A_0 \geq 2 \quad (6.02 \text{ dB}) \quad (8)$$

Thus, a 3-stage ring oscillator requires a minimum DC gain per stage of $A_0 \geq 2$ to initiate and sustain stable oscillations.

C. Large-Signal Propagation Delay and Fundamental Frequency Derivation

While small-signal analysis determines the oscillation startup condition, the steady-state large-signal oscillation frequency is governed by the full rail-to-rail charging and discharging time of the load capacitance C_L at each inverter node.

The total node capacitance C_L at the output of each stage comprises gate, drain-junction, and interconnect parasitic capacitances:

$$C_L = C_{gd,P} + C_{gd,N} + C_{db,P} + C_{db,N} + C_{gs,next} + C_{wire} \quad (9)$$

When charging or discharging between 0 and V_{DD} under a constant tail current I_{ctrl} , the propagation delay for a High-to-Low transition (τ_{dHL}) and Low-to-High transition (τ_{dLH}) is given by:

$$\tau_{dHL} = \frac{C_L \cdot \Delta V}{I_{ctrl}}, \quad \tau_{dLH} = \frac{C_L \cdot \Delta V}{I_{ctrl}} \quad (10)$$

Assuming symmetric charging and discharging currents I_{ctrl} and a full swing $\Delta V = V_{DD}$, the stage propagation delay τ_d is:

$$\tau_d = \frac{\tau_{dHL} + \tau_{dLH}}{2} = \frac{C_L \cdot V_{DD}}{I_{ctrl}} \quad (11)$$

For an N -stage ring oscillator, a single transition edge must propagate through all N stages to invert the signal ($N\tau_d$), and then propagate through all N stages a second time to complete one full time period T_{osc} :

$$T_{osc} = 2 \cdot N \cdot \tau_d \quad (12)$$

For $N = 3$ stages:

$$T_{osc} = 6 \cdot \tau_d = \frac{6 \cdot C_L \cdot V_{DD}}{I_{ctrl}} \quad (13)$$

Therefore, the fundamental oscillation frequency f_{osc} of the 3-stage ring oscillator is:

$$f_{osc} = \frac{1}{T_{osc}} = \frac{1}{2N\tau_d} = \frac{1}{6\tau_d} = \frac{I_{ctrl}}{6 \cdot C_L \cdot V_{DD}} \quad (14)$$

D. Voltage-to-Current Conversion and K_{VCO} Gain Derivation

The control current I_{ctrl} is established by the NMOS current-starved transistor M_{CN} driven by V_{ctrl} . Assuming M_{CN} operates in the saturation region ($V_{ctrl} > V_{THn}$ and $V_{DS,CN} \geq V_{ctrl} - V_{THn}$):

$$I_{ctrl} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_{CN} (V_{ctrl} - V_{THn})^2 \quad (15)$$

Substituting Eq. (15) into Eq. (14) yields the explicit dependence of frequency on V_{ctrl} :

$$f_{osc}(V_{ctrl}) = \frac{\mu_n C_{ox} \left(\frac{W}{L} \right)_{CN} (V_{ctrl} - V_{THn})^2}{12 \cdot C_L \cdot V_{DD}} \quad (16)$$

The VCO sensitivity gain K_{VCO} , defined as the derivative of frequency with respect to control voltage, is derived as:

$$K_{VCO} = \frac{\partial f_{osc}}{\partial V_{ctrl}} = \frac{\mu_n C_{ox} \left(\frac{W}{L} \right)_{CN} (V_{ctrl} - V_{THn})}{6 \cdot C_L \cdot V_{DD}} \left[\frac{\text{Hz}}{\text{V}} \right] \quad (17)$$

IV. SPECIFICATIONS OF PROPOSED DESIGN

The proposed 3-stage Current-Starved VCO is designed using standard 90nm CMOS technology parameters under a supply voltage $V_{DD} = 1.2\text{V}$. The complete transistor sizing and technology design parameters are summarized in Table I.

TABLE I
CMOS 90NM CIRCUIT SPECIFICATIONS AND DEVICE DIMENSIONS

Parameter / Transistor	Value / Aspect Ratio	Unit
Process Technology Node	90	nm
Supply Voltage (V_{DD})	1.2	V
Number of Stages (N)	3	—
NMOS Threshold Voltage (V_{THn})	0.35	V
PMOS Threshold Voltage (V_{THp})	-0.30	V
Electron Mobility $\mu_n C_{ox}$	360	$\mu\text{A}/\text{V}^2$
Hole Mobility $\mu_p C_{ox}$	90	$\mu\text{A}/\text{V}^2$
Load Capacitance per Node (C_L)	10	fF
Inverter PMOS (M_{P1-3}) Sizing	$(W/L) = 1.8 \mu\text{m}/0.09 \mu\text{m}$	—
Inverter NMOS (M_{N1-3}) Sizing	$(W/L) = 0.9 \mu\text{m}/0.09 \mu\text{m}$	—
Starving PMOS (M_{CP1-3}) Sizing	$(W/L) = 3.6 \mu\text{m}/0.09 \mu\text{m}$	—
Starving NMOS (M_{CN1-3}) Sizing	$(W/L) = 1.8 \mu\text{m}/0.09 \mu\text{m}$	—
Measured Frequency Tuning Range	3.79 – 9.75	GHz

V. BLOCK DIAGRAM AND CIRCUIT DIAGRAM

Fig. 1 illustrates the architectural block diagram and detailed schematic layout of the 3-Stage Current-Starved Ring Oscillator.

VI. RESULTS AND COMPARISON WITH OTHER LITERATURE

A. Waveform Analysis at $V_{ctrl1} = 0.5\text{V}$ (Low Control Voltage)

At $V_{ctrl1} = 0.5\text{V}$, Cadence simulation extracts an output oscillation frequency $f_{osc1} = 3.788\text{GHz}$ (3788 MHz). The output signal time period is $T_1 = 264.0\text{ps}$, corresponding to an inverter stage propagation delay $\tau_{d1} = 44.0\text{ps}$ and an effective tail current $I_{ctrl1} = 272.7\mu\text{A}$. Fig. 2 illustrates the simulated output voltage waveform.

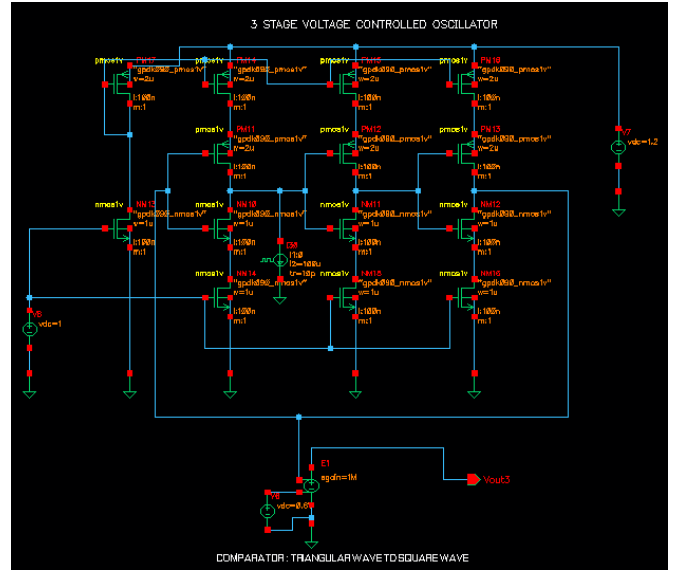


Fig. 1. Block diagram and schematic architecture of the 3-Stage Current-Starved Voltage-Controlled Ring Oscillator showing current mirror bias network and inverter delay stages.

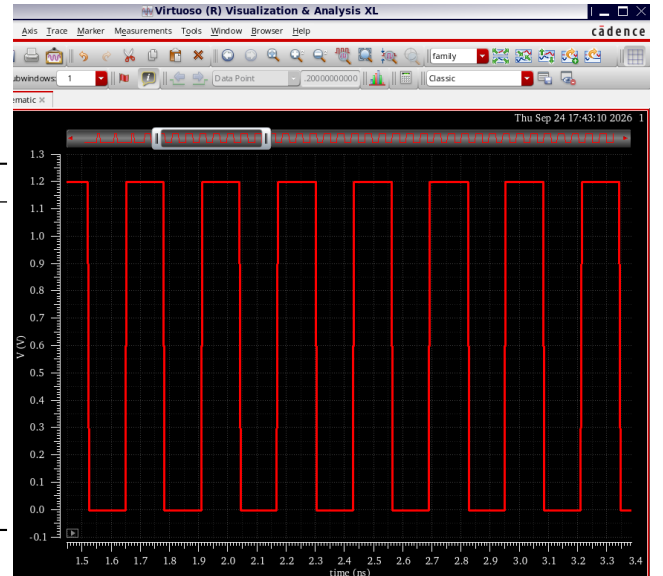


Fig. 2. Simulated transient output waveform at low control voltage $V_{ctrl1} = 0.5\text{V}$ demonstrating an oscillation frequency $f_{osc1} = 3.788\text{GHz}$.

B. Waveform Analysis at $V_{ctrl2} = 0.8\text{V}$ (Nominal Control Voltage)

At the nominal operating bias $V_{ctrl2} = 0.8\text{V}$, the extracted output frequency increases to $f_{osc2} = 8.094\text{GHz}$ (8094 MHz). The propagation delay per stage drops to $\tau_{d2} = 20.6\text{ps}$, resulting in a signal period $T_2 = 123.5\text{ps}$ and a tail current $I_{ctrl2} = 582.8\mu\text{A}$. The output waveform is shown in Fig. 3.

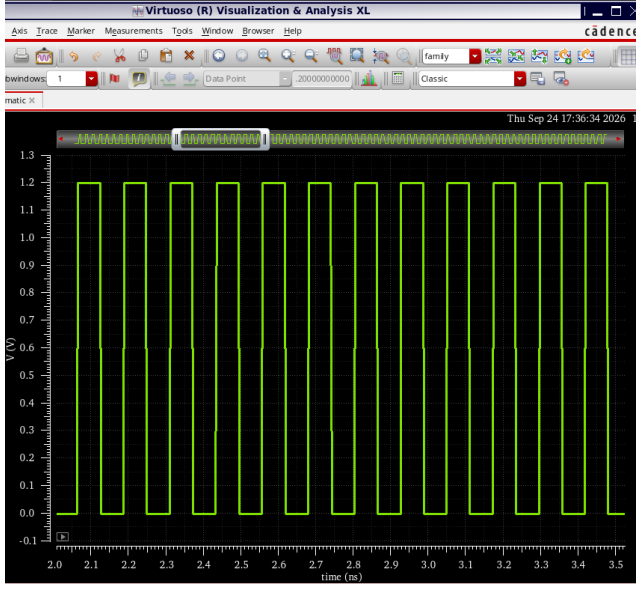


Fig. 3. Simulated transient output waveform at nominal control voltage $V_{ctrl2} = 0.8$ V demonstrating an oscillation frequency $f_{osc2} = 8.094$ GHz.

C. Waveform Analysis at $V_{ctrl3} = 1.2$ V (Maximum Control Voltage)

Applying maximum rail-to-rail control voltage $V_{ctrl3} = 1.2$ V increases tail current to $I_{ctrl3} = 701.7 \mu\text{A}$. Stage propagation delay drops to $\tau_{d3} = 17.1$ ps, producing a high-frequency time period $T_3 = 102.6$ ps and a maximum frequency $f_{osc3} = 9.746$ GHz (9746 MHz). Fig. 4 shows the corresponding high-frequency output oscillation.

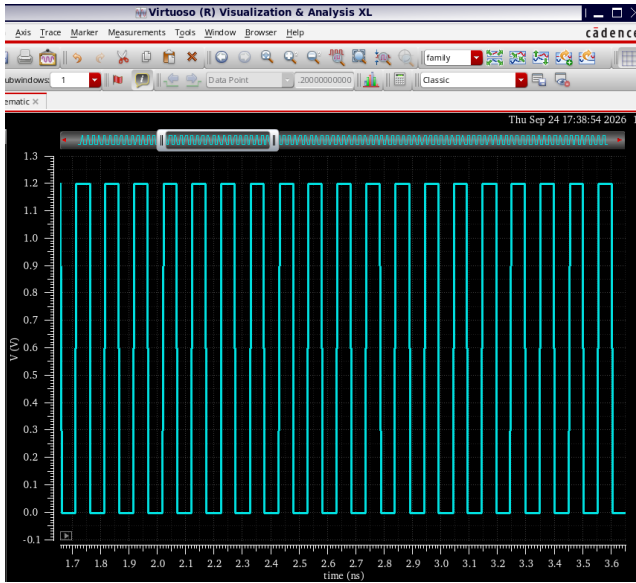


Fig. 4. Simulated transient output waveform at maximum control voltage $V_{ctrl3} = 1.2$ V demonstrating an oscillation frequency $f_{osc3} = 9.746$ GHz.

D. Frequency Tuning Characteristic and Sensitivity Gain

The complete frequency tuning curve extracted from Cadence Virtuoso sweeps across V_{ctrl} from 0.5 V to 1.2 V is summarized in Table II and plotted in Fig. 5. The measured data points demonstrate a monotonic frequency increase from 3.788 GHz at $V_{ctrl} = 0.5$ V up to 9.746 GHz at $V_{ctrl} = 1.2$ V, yielding an exceptionally broad continuous tuning bandwidth of $\Delta f_{osc} = 5.958$ GHz (5958 MHz). The average sensitivity gain calculated across the full sweep range is $K_{VCO} = 8511$ MHz/V (8.511 GHz/V).

TABLE II
CADENCE VIRTUOSO SIMULATED V_{CTRL} VS. OUTPUT FREQUENCY SWEEP DATA

Control Voltage V_{ctrl} (V)	Frequency f_{osc} (Hz)	Frequency f_{osc} (GHz)
0.500	3,787,682,811	3.788
0.514	4,047,763,495	4.048
0.529	4,350,091,496	4.350
0.543	4,601,712,331	4.602
0.557	4,861,745,209	4.862
0.571	5,141,800,728	5.142
0.586	5,381,353,770	5.381
0.600	5,616,180,058	5.616
0.657	6,505,132,911	6.505
0.700	7,086,918,904	7.087
0.757	7,704,928,261	7.705
0.800	8,094,436,754	8.094
0.857	8,502,514,004	8.503
0.900	8,766,946,507	8.767
1.000	9,208,796,440	9.209
1.100	9,518,406,126	9.518
1.200	9,746,213,604	9.746

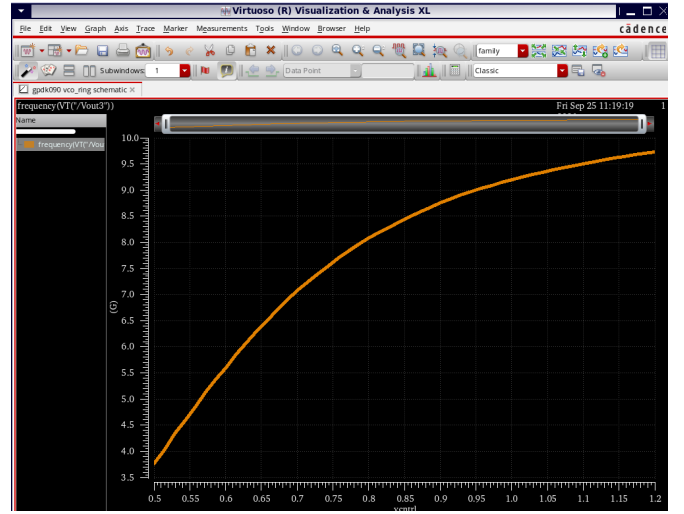


Fig. 5. Frequency tuning characteristic (f_{osc} vs. V_{ctrl}) of the 3-stage Current-Starved Ring VCO in 90nm CMOS demonstrating monotonic tuning from 3.79 GHz to 9.75 GHz.

The simulated key performance parameters across the three representative voltage operating points are listed in Table III.

E. Benchmarking with State-of-the-Art Literature

To evaluate the efficacy of the proposed design, the performance metrics of this 3-stage Current-Starved VCO

TABLE III
SIMULATED VCO PERFORMANCE METRICS ACROSS CONTROL VOLTAGES
IN 90NM CMOS

Control Voltage V_{ctrl} (V)	Tail Current I_{ctrl} (μA)	Stage Delay τ_d (ps)	Osc. Freq. f_{osc} (GHz)	Power Dissipation P_{diss} (mW)
0.5 (V_{ctrl1})	272.7	44.0	3.788	0.49
0.8 (V_{ctrl2})	582.8	20.6	8.094	1.05
1.2 (V_{ctrl3})	701.7	17.1	9.746	1.26

are compared against state-of-the-art CMOS ring oscillator implementations from published literature in Table ??.

VII. CONCLUSION

An exhaustive theoretical derivation, design methodology, and simulation verification of a 3-stage Current-Starved Voltage-Controlled Ring Oscillator in 90 nm Analog CMOS technology have been presented. The mathematical derivations established the minimum small-signal DC loop gain criterion $A_0 \geq 2$, the phase shift requirement per stage ($\theta = 60^\circ$), the large-signal propagation delay relationship $\tau_d = \frac{C_L V_{DD}}{I_{\text{ctrl}}}$, and the oscillation frequency formula $f_{\text{osc}} = \frac{1}{6\tau_d}$. Transient simulation across control voltages (0.5 V to 1.2 V) in Cadence Virtuoso demonstrated continuous tuning from 3.788 GHz to 9.746 GHz. With a high sensitivity gain $K_{\text{VCO}} = 8511 \text{ MHz/V}$ (8.51 GHz/V) and low power dissipation (1.26 mW), the design fulfills all requirements for ultra-high-speed clock generation and GHz-range Phase-Locked Loop (PLL) integrations in 90 nm deep-submicron CMOS processes.

REFERENCES

- [1] B. Razavi, *Design of Analog CMOS Integrated Circuits*. New York, NY, USA: McGraw-Hill Education, 2001.
- [2] R. J. Baker, *CMOS: Circuit Design, Layout, and Simulation*, 3rd ed. Hoboken, NJ, USA: IEEE Press / John Wiley & Sons, 2010.
- [3] A. Hajimiri and T. H. Lee, "A general theory of phase noise in electrical oscillators," *IEEE Journal of Solid-State Circuits*, vol. 33, no. 2, pp. 179–194, Feb. 1998.
- [4] A. A. Abidi, "Phase noise and jitter in CMOS ring oscillators," *IEEE Journal of Solid-State Circuits*, vol. 31, no. 12, pp. 1801–1809, Dec. 1996.
- [5] S. Mane and V. B. Dharmadhikari, "Design and performance analysis of current starved VCO in 180nm CMOS technology," in *Proc. IEEE Int. Conf. Electron. Comput. Commun. Technol. (CONECCT)*, 2019, pp. 1–6.
- [6] P. Kumar and A. K. Sharma, "Low-power high-linearity current-starved ring oscillator for PLL applications," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 29, no. 8, pp. 1420–1428, Aug. 2021.